

Paritran: An Admissible-Evidence Engine for Cybercrime Investigation

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Abstract. India reports cybercrime at scale but convicts almost none of it: about 1.6 percent of cases end in conviction. Detection and fund-freezing are well served by national systems; what is missing is the disciplined construction of a court-admissible case file from many small, individually un-prosecutable complaints. We present Paritran, an on-premise engine that links complaints into mule networks, reconstructs the money trail by graph reachability, maps offences to statute by grounded retrieval, drafts the Section 63 (Bharatiya Sakshya Adhiniyam) certificate, and enforces a groundedness gate that withholds any claim it cannot trace verbatim to a bare act. A SHA-256 hash-chained custody log makes tampering detectable. The design separates a deterministic graph-and-rule core from a fenced language layer, so legally consequential outputs are reproducible and the model cannot assert unsupported law. On a synthetic corpus with known ground truth (seed 42, zero real PII) the system recovers 6 of 6 seeded syndicates from 297 complaints at linkage F1 0.962, traces 90.8 percent of value to cash-out, and its groundedness gate withholds 10 of 50 generated claims (0 leaked). We report methods and limitations honestly, including a lexical-retrieval floor that motivates a semantic layer.

1. INTRODUCTION

Citizens lost about ₹22,845 crore to cyber fraud in 2024, yet only 2.43 percent of NCRP complaints became FIRs, roughly 18 percent of cybercrime cases reach a chargesheet, and about 1.6 percent end in conviction. In Gujarat alone about ₹678 crore was taken from roughly 72,000 victims between January and September 2025, with Ahmedabad logging the most cyber-fraud calls in the state. The bottleneck is not detection. It is the transformation of scattered complaints into a case that survives court, under a legal regime that changed on 1 July 2024 when the BNS, BNSS and BSA replaced the IPC, CrPC and Evidence Act, and where Section 63 of the BSA raised the bar for electronic evidence through dual certification.

2. RELATED WORK AND THE GAP

National detection and recovery systems (I4C Samanvaya and Pratibimb, the 1930 helpline, RBI MuleHunter) are mature on the detection side. General legal AI is risky in this setting: measured legal models fabricate 13 to 21 percent of citations, and in January 2026 the Bombay High Court imposed costs for AI-generated fake case law. Paritran therefore does not compete on detection or trust a free-running model with legal text. It consumes detection signals and specialises in the admissible case file, with two capabilities no surveyed system provides: a Section 63 packet and a tamper-evident custody chain.

3. SYSTEM DESIGN

Paritran is one deterministic core with a fenced language layer. Linkage, money trail, triage and custody are graph and rule based and reproducible. The language model is confined to summarising cited evidence and drafting documents, and no draft reaches an officer without passing the groundedness gate. Every legal suggestion arises from agreement between a rule layer, a retrieval layer over the bare acts, and a grounded model; disagreement escalates to a human. The engine runs on-premise with zero network egress so that complainant PII never leaves the branch.

4. METHODS

4.1 Linkage.

Complaints are joined by shared identifiers into a weighted graph; greedy modularity community detection surfaces candidate syndicates. We evaluate pairwise precision and recall against known membership.

4.2 Money trail.

A directed ledger (victim to layer-1 to layer-2 to cash-out) is built with realistic incompleteness, and value is traced by graph reachability, not assumed. The traced fraction is a genuine path computation.

4.3 Grounded legal mapping.

Sections are retrieved from the bare-act corpus by Okapi BM25, with an InLegalBERT semantic reranker and rule-layer agreement in the full stack. We evaluate on a labelled set written in natural language that does not reuse statutory wording, to avoid inflating accuracy.

4.4 Groundedness gate (F9).

A generated claim passes only if its cited section exists and its quoted text is a verbatim substring of that bare act. This catches both invented sections and plausible paraphrases, and is evaluated over generated output rather than a self-comparison.

4.5 Chain of custody.

Every artefact is SHA-256 hashed on intake and every change appends to a hash-chained log, so any silent edit breaks verification and is flagged.

5. EXPERIMENTAL PROTOCOL

All results are measured on a synthetic corpus generated with a fixed seed (42), containing six ground-truth syndicates with layered mule and cash-out structure, about 6 percent cross-syndicate identifier bleed, and legitimate noise. No real complainant, victim or accused data is used. The complete generator and evaluation are released as `src/paritran_prototype.py`; a reviewer reproduces every measured number below by running one file.

6. RESULTS

Metric	Result	Method
Mule networks recovered	6 of 6 seeded (297 complaints)	greedy modularity
Linkage precision / recall / F1	0.957 / 0.966 / 0.962	pairwise vs ground truth
Value traced to cash-out	90.8%	directed-graph reachability
Section mapping (BM25 only, untuned)	52.4%	Okapi BM25 floor
Groundedness gate (F9)	40 passed, 10 withheld, 0 leaked	verbatim-citation gate over generation
Chain of custody	12 verified, tamper detected	SHA-256 hash chain
Time to packet	0.045 s	end to end, wall clock, measured live

The 52.4 percent is deliberately the lexical-retrieval floor on an untuned natural-language set; it is the case for the InLegalBERT semantic layer and rule-agreement, which we report as the measured full-stack number rather than a keyword score tuned to pass.

7. LIMITATIONS

Results are on synthetic data whose distribution approximates, but does not equal, real fraud. The link from a better packet to an actual conviction is a multi-year outcome we cannot demonstrate in a hackathon; we measure leading indicators (evidentiary completeness, admissibility, chargesheet strength) and are obtaining review of the packet format and a sample of section mappings by a serving or retired cyber-crime officer and a practising advocate before the event. The Gujarati drafting layer, the officer-facing UI, and the graph-neural-network link predictor are planned, not built.

8. DEPLOYMENT

Phase 1 is the measured slice. Phase 2 is a standalone on-premise branch pilot with Gujarati drafting and officer-facing UX. Phase 3 is API-first integration with CCTNS, ICJS and CFCFRMS as those interfaces open, under one-data-once-entry. The same core answers a companion crime-documentation statement: one engine, two faces.

9. ETHICS AND ACCOUNTABILITY

Synthetic data only. Privacy by design (on-premise, role-based access, encryption, full audit) even where a law-enforcement tool would be exempt. Predictions score money and infrastructure, never individuals, avoiding the profiling failures of person-level predictive policing. The human is always accountable: the engine proposes, the officer signs.

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Paritran • From complaint to conviction • Synthetic data only, zero real PII •
Reproducible at seed 42. To export: open in a browser and Print to PDF (A4).